Surrogate-conditioned optimizer ranking in block-scale urban energy design: A comparison of policy learning and Pareto search

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Abstract

Block-scale urban energy design requires optimization methods that connect urban form with building energy demand, rooftop photovoltaic potential, and winter solar access during early-stage planning. This study develops a surrogate-conditioned framework to compare policy learning and Pareto search for residential-block morphology optimization. Feasible block layouts are generated first and converted into 12 morphology descriptors. A morphology-to-performance response surface then assigns three targets to each sample: Energy Use Intensity, Energy Generation, and winter sunlight hours. These targets are denoted EUIt, EG, and H throughout the paper. The 2000-sample DNN surrogate selected from nested 500/1000/1500/2000-sample pools provides the common environment for DDPG and NSGA-II. Repeated five-fold validation gives mean nMAE values of 0.0176, 0.0265, and 0.0145 for EUIt, EG, and H. Under the shared surrogate and matched query budget, NSGA-II attains near-ceiling descriptor-space HV, while DDPG shows preference-dependent performance. Projection to generated feasible blocks compresses the 2000-candidate NSGA-II archive to 51 unique feasible blocks, indicating that descriptor-space coverage and feasible-morphology diversity differ substantially. A 24-case cross-model stress test completes all locked cases and shows limited metric and rank consistency between analytic and physics-based outputs. A four-block cross-climate sensitivity analysis shows stable EG ordering but climate-dependent EUIt and H responses. The study contributes a representation-aware urban design workflow that links surrogate search, feasible-block projection, and sensitivity analysis to improve the interpretability of optimizer comparisons for early-stage urban energy planning.

Keywords: Surrogate-assisted optimization, Urban design, Urban morphology, Deep reinforcement learning, Multi-objective optimization

Nomenclature

Abbreviations

EUIt	Energy Use Intensity	EG	Energy Generation
H	winter sunlight hours	DNN	deep neural network
DDPG	Deep Deterministic Policy Gradient	NSGA-II	Non-dominated Sorting Genetic Algorithm II
CMA-ES	Covariance Matrix Adaptation Evolution Strategy	GHI	global horizontal irradiance
HV	Hypervolume	IGD	Inverted Generational Distance
nMAE	normalized mean absolute error	OSLI	open-space location index

Formula symbols

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$\mathbf{x}$	morphology descriptor vector	$\mathbf{y}^{ana}$	analytic target vector
y_i	target value for objective i	z_i	normalized target coordinate
$y_i^{\min}$	lower target bound	$y_i^{\max}$	upper target bound
$\mathbf{u}$	ideal vector $[0, 1, 1]^{\mathrm{T}}$	$\mathbf{w}$	axis-scaling coefficient vector
d_w	weighted normalized distance	R	DDPG training reward
θ	block orientation angle	ϵ	response-generation noise term
A_{gross}	gross floor area	A_{block}	block area
A_{foot}	building footprint area	A_{open}	open-space area
A_{env}	envelope area	P_{sum}	summed footprint perimeter
$\bar{h}$	mean building height	$h_{\max}$	maximum building height
W_{ew}	east-west canyon width	W_{ns}	north-south canyon width

1. Introduction

1.1. Motivation

Urban energy planning is increasingly shaped by the dual requirement of reducing building-sector emissions and maintaining reliable energy services. At the district and block scales, urban morphology controls floor area, envelope exposure, roof availability, obstruction, and solar access, which together affect both energy demand and distributed generation potential [1, 2]. These spatial effects are important for Chinese urban districts, where renewal, photovoltaic deployment, and climate-resilient planning often need to be assessed before detailed building-system design is available [3, 4].

Block morphology influences energy performance through coupled physical mechanisms. Density and height alter heating and cooling demand by changing envelope exposure, mutual shading, and microclimatic context [5, 6]. Solar access and daylight availability depend on roof exposure, sky-view conditions, block orientation, and open-space arrangement [7, 8]. Recent large-scale studies further show that urban spatial structure can shape building energy demand across broad urban areas and under extreme climate events, which makes morphology-sensitive design relevant beyond isolated building cases [9, 10].

The computational challenge is that morphology-sensitive energy analysis becomes expensive when thousands of candidate blocks, multiple optimizer seeds, and several preference settings must be evaluated. Simulation-based urban form generation and building-performance optimization provide a strong basis for this task, but repeated full physical simulation remains costly in search loops [11, 12]. Surrogate-assisted optimization offers a practical screening layer by replacing repeated expensive evaluations with a trained response model during optimization. This acceleration changes the interpretation of the optimizer comparison: the resulting method ordering depends not only on the optimizer, but also on descriptor coverage, surrogate selection, retained outputs, and the mapping from continuous descriptor candidates to generated feasible blocks.

Accordingly, this study evaluates policy learning and Pareto search within a shared block-morphology workflow. The optimizer comparison is first performed on an explicit morphology-to-performance response surface that assigns EUIt, EG, and H to generated feasible blocks and provides the target values for surrogate training. The resulting descriptor-space candidates are then mapped to generated feasible blocks and examined in selected physics-based and climate-sensitive cases. This sequence separates three questions: how the optimizers behave on the surrogate, how much their outputs compress after feasible-block projection, and how selected blocks respond under cross-model and climate-sensitive calculations.

1.2. Literature review

Simulation-based optimization has been widely used to connect energy performance with design variables in building and urban studies. General reviews establish the role of optimization algorithms in building performance analysis and show that conclusions depend on model assumptions, objective definitions, and search protocols [13, 14]. Surrogate modelling has expanded this field by reducing evaluation cost and enabling larger design sweeps, while coordinated surrogate studies demonstrate its value for coupled building and microclimate problems [15, 16]. Recent district-scale morphology optimization work further shows that surrogate-assisted frameworks can reduce the computational cost of exploring morphology effects on building energy demand [17]. These studies provide the computational basis for the present work, but they leave open how method ordering should be interpreted when the comparison itself is conditioned by a surrogate response surface.

Parametric design studies show that geometry generation can support energy- and daylight-oriented design exploration. Research on parametric daylighting and low-energy design demonstrates that early-stage geometry and envelope choices can be formalized as searchable design spaces [18, 19]. Evolutionary multi-objective optimization remains widely used because it provides explicit trade-off archives for conflicting objectives. NSGA-II and related methods have been applied to envelope design, photovoltaic-oriented inverse urban design, urban block optimization, and energy-efficient shading design [20, 21]. Additional studies extend this logic to urban heat-island effects, retrofitting, cross-building optimization, adaptive BIPV shading, and shape-envelope design [22, 23]. These works demonstrate the value of Pareto search for design exploration, while also showing that an archive must ultimately be interpreted in relation to feasible building or block forms [24, 25]. In morphology optimization, this issue becomes more pronounced because continuous descriptor outputs may not correspond one-to-one to generated feasible blocks [26, 27]. Shape-envelope optimization studies raise a related point: objective-space performance needs to be connected back to physically meaningful form decisions [28].

Deep reinforcement learning provides another family of search methods, but its role in morphology design requires careful framing. Sequential decision problems are commonly described through Markov Decision Process formulations in reinforcement learning [30, 31]. However, the morphology task in this study is not treated as a physical-time design-control process. DDPG is used as a serialized static black-box search method on a guarded surrogate response surface. General DRL reviews and foundational DQN and DDPG studies establish policy learning for sequential decision-making and continuous control [29, 32]. Continuous-control applications in robotics and games illustrate the breadth of actor-critic learning beyond energy design [33, 34]. Reinforcement learning has also been used in visual captioning and human-machine cooperative driving, showing that policy-learning formulations can be adapted to tasks with different state and action structures [35, 36]. In the present study, DDPG is adopted as a continuous-action policy-learning comparator because it provides a clear actor-critic mechanism for generating descriptor queries in a bounded surrogate environment [37].

In the energy domain, DRL has been widely adopted for HVAC control, building operation, distribution-network control, and renewable-building operation [38, 39]. These applications are primarily operational control problems in which states evolve through time. Static urban morphology optimization differs from that setting because each query is an absolute candidate description evaluated by a surrogate response surface. Recent work on distribution-network control and active distribution networks further shows that energy-system optimization may include grid constraints beyond the block morphology objectives considered here [40, 41]. Experimental HVAC-control studies and automated DRL scheduling frameworks examine stronger or more adaptive actor-critic alternatives under operational objectives [42, 43]. Derivative-free and hybrid energy-system methods remain important neighbouring baselines because renewable-building operation and photovoltaic-hydrogen sizing can require continuous search without a learned policy [44, 45]. The present work therefore uses DDPG as a policy-learning comparator and treats broader continuous-optimizer coverage as an extension for future studies.

The remaining gap is methodological and representation-oriented. Urban energy optimization has established the value of morphology-aware design, surrogate modelling has reduced the cost of repeated evaluation, and reinforcement learning has proved useful in several energy-operation settings. What remains less clear is how policy learning and Pareto search should be compared when both operate on a shared surrogate response surface, when the surrogate inputs are descriptors derived from generated blocks, and when continuous outputs require projection before they can be interpreted as feasible morphologies. This gap matters for early-stage design support because a method can appear strong in descriptor space while mapping to a smaller feasible-block set or showing limited rank consistency under physics-based and climate-sensitive calculations.

1.3. Contributions and paper organization

This study develops a surrogate-assisted comparison workflow for block-scale urban energy design. The contributions are threefold.

1. The study establishes a morphology-to-performance workflow in which feasible residential block layouts are generated before 12 morphology descriptors are calculated, and in which EUIt, EG, and H are assigned by an explicit analytic response surface.

2. The study compares DDPG and NSGA-II under the same selected DNN surrogate and matched query budget, while separating descriptor-space outputs, equal-size retained outputs, and nearest-feasible-block projections.

3. The study connects surrogate-based optimization with repeated surrogate assessment, feasible-block projection, a 24-case physics-based cross-model stress test, and a four-block cross-climate sensitivity analysis, thereby clarifying the conditions under which optimizer comparisons can support early-stage design interpretation.

The remainder of the paper is organized as follows. Section 2 presents the morphology generation, analytic response assignment, surrogate model, optimizer protocol, feasible projection, and sensitivity analyses. Section 3 presents the surrogate, optimizer, projection, cross-model, and climate results. Section 4 discusses implications and limitations. Section 5 concludes the paper. Detailed equations, additional optimizer results, case-level physical outputs, and climate tables are provided in the Supplementary Information.

2. Methodology

The methodology links morphology generation, analytic target assignment, surrogate-assisted optimization, and post-optimization assessment within one block-scale design workflow. The workflow distinguishes three representations: generated feasible block geometries, 12 morphology descriptors used by the surrogate and optimizers, and locked block cases examined through physics-based and climate-sensitive calculations. This distinction is essential because the optimizer operates in descriptor space, whereas design interpretation ultimately depends on generated block geometries. Figure 1 summarizes the computational sequence.

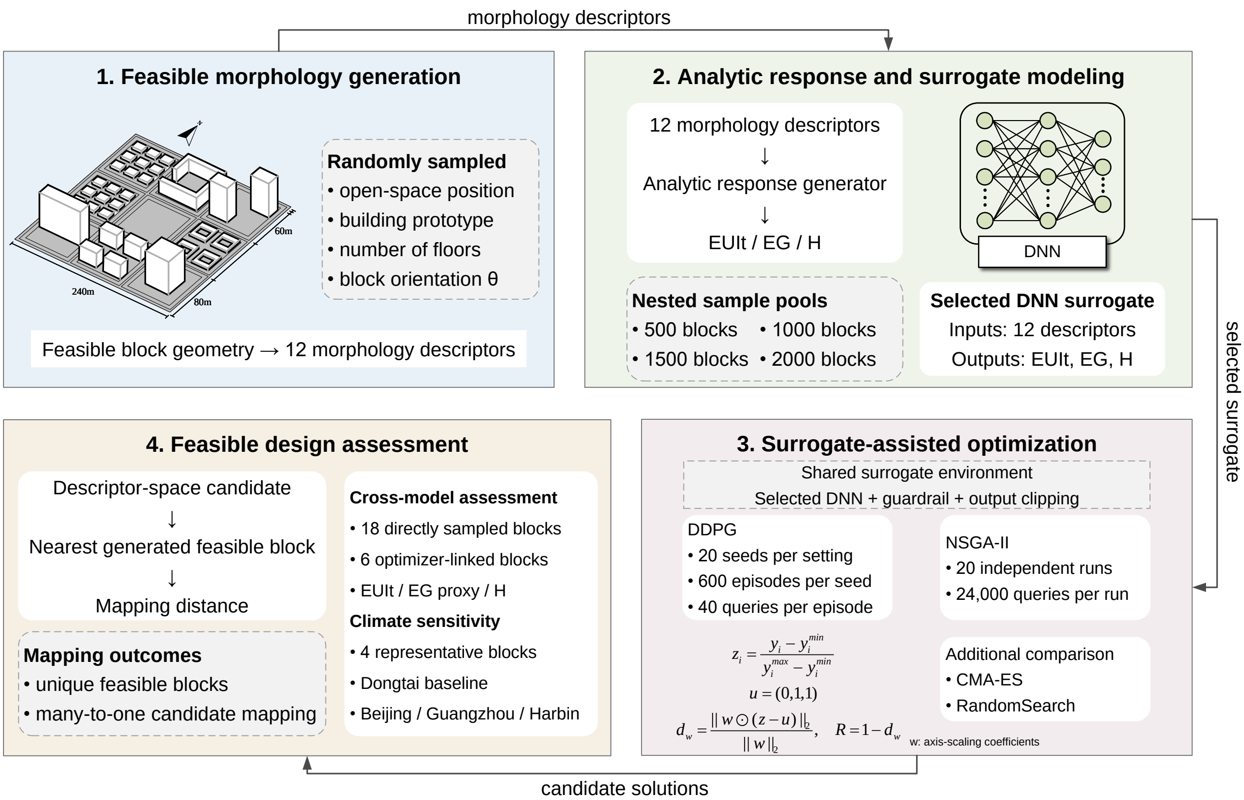

Figure 1: Workflow of the surrogate-assisted block-scale morphology optimization framework. Feasible block layouts are generated and converted into 12 morphology descriptors. The analytic response and surrogate-modelling stage assigns EUIt, EG, and H and trains the selected DNN surrogate. DDPG and NSGA-II are compared under the shared surrogate, then descriptor-space candidates are projected to feasible blocks for selected physical and climate-sensitivity checks.

2.1. *Feasible morphology generation and descriptor construction*

The morphology generator first constructs feasible block layouts and only then calculates descriptor values. Each block is represented as a 240 m × 240 m square divided into a 3 × 3 grid of 80 m × 80 m land units. One cell is assigned as open space, and the remaining eight cells are populated with residential prototypes. For each generated block, the generator samples the open-space position, building prototype, number of floors, and orientation angle. The orientation angle is sampled within -45 deg to 45 deg. The resulting block is therefore a generated feasible morphology, while the 12 input quantities used by the surrogate are descriptors derived from that morphology.

The descriptor set in Table 1 includes FAR, SD, AF, AR_ew, AR_ns, SVF, BD, OSR, SC, PAR, theta, and OSLI. These quantities are derived morphology descriptors rather than independently sampled design variables. For example, FAR, BD, AF, and OSR are linked through the same footprint and gross-floor-area quantities, while OSLI is an integer descriptor of open-space position. Consequently, optimizer outputs are first interpreted in descriptor space and then mapped to generated feasible blocks before morphology-level conclusions are drawn.

Table 1: Morphology descriptors used by the surrogate and optimizer interface.

Descriptor	Unit	Calculation	Note
FAR	–	$A_{\text{gross}}/A_{\text{block}}$	density
SD	m	$h_{\max} - \bar{h}$	height variation
AF	floors	$A_{\text{gross}}/A_{\text{foot}}$	linked to FAR, BD
AR_{ew}	–	$\bar{h}/W_{ew}$	east-west canyon
AR_{ns}	–	$\bar{h}/W_{ns}$	north-south canyon
SVF	–	analytic openness proxy	sky-view openness
BD	–	$A_{\text{foot}}/A_{\text{block}}$	footprint density
OSR	–	$A_{\text{open}}/A_{\text{gross}}$	open-space ratio
SC	–	$A_{\text{env}}/A_{\text{gross}}$	envelope exposure
PAR	m^{-1}	$P_{\text{sum}}/A_{\text{foot}}$	compactness
θ	deg	sampled in $[-45, 45]$	orientation
OSLI	integer	0–8	open-space cell

2.2. *Analytic response generation*

In this study, the analytic response generator is the target-assignment module that maps a generated morphology descriptor vector, denoted $\mathbf{x}$, to three target responses, denoted $\mathbf{y}^{ana} = [\text{EUIt}, \text{EG}, H]^{\mathsf{T}}$. The mapping uses explicit formulas that combine density, openness, aspect ratio, sky-view factor, orientation, and open-space position to represent demand, rooftop-generation, and sunlight-access responses. A small stochastic term is added during dataset construction, and the target values are then limited to the response ranges used for surrogate training.

This module provides a controlled morphology-to-performance response surface for optimizer comparison. The nested 500/1000/1500/2000-sample pools are deterministic subsets of the same generated morphology pool under the recorded random seed and morphology-generation protocol. The physics-based case calculations are conducted subsequently on locked generated blocks to assess how the analytic target values relate to EnergyPlus- and Radiance-component outputs in selected cases.

The response directions follow the design objectives: lower EUIt is preferred, while higher EG and H are preferred. The active 2000-sample pool is the largest nested sample regime used for the selected DNN surrogate. The full equations, station-bias convention, stochastic terms, and clipping ranges are given in Supplementary Section S1. Table 2 distinguishes analytic response assignment, surrogate prediction, feasible projection, and the post-optimization sensitivity analyses.

2.3. *Surrogate model training and robustness assessment*

The selected surrogate is assessed through repeated five-fold cross-validation, leave-one-OSLI-out validation, outer-shell holdout, and feature-tail holdout. The repeated five-fold protocol evaluates interpolation performance under randomized folds. The leave-one-OSLI-out protocol tests whether the surrogate remains stable when one open-space-position class is withheld. The outer-shell and feature-tail holdouts examine surrogate behaviour near the boundary of the sampled descriptor domain.

Table 2: Evaluation modes in the study workflow.

Mode	Source	Output	Use	Limit
Analytic response generation	generated descriptors	EUIt, EG, H	target assignment	analytic surface
Surrogate prediction	12 descriptors	DNN outputs	optimizer calls	selected checkpoint
Feasible projection	descriptor candidate	nearest block	morphology mapping	nearest neighbour
Physical stress test	locked blocks	physical outputs	cross-model check	24 cases
Climate sensitivity	four blocks	weather deltas	climate response	no retraining

The selected DNN comes from the 2000-sample regime of the nested scale study. Two complementary assessments are used. The scale study selects the 2000-sample surrogate used for optimization, while the robustness protocols quantify how consistently that surrogate reproduces the analytic target surface across random folds, OSLI groups, and boundary-oriented holdouts. The repeated cross-validation mean nMAE values are 0.0176 for EUIt, 0.0265 for EG, and 0.0145 for H, with corresponding Spearman correlations of 0.991, 0.984, and 0.995.

Table 3: Surrogate robustness across analytic-target validation regimes.

Validation regime	nMAE			Spearman ρ		
	EUIt	EG	H	EUIt	EG	H
Repeated 5×5 CV	0.0176	0.0265	0.0145	0.991	0.984	0.995
Leave-one-OSLI-out	0.0185	0.0342	0.0185	0.990	0.972	0.992
Outer-shell holdout	0.0278	0.0335	0.0208	0.985	0.975	0.994
Feature-tail holdout	0.0230	0.0287	0.0198	0.986	0.983	0.992

nMAE and Spearman ρ compare DNN predictions with analytic target values assigned to the generated morphology samples.

2.4. *DDPG as serialized static black-box search*

The DDPG environment serializes repeated queries to a static guarded surrogate. At the beginning of each episode, a random descriptor query is evaluated to construct the initial normalized three-target state. At each subsequent step, the actor outputs an absolute 12-dimensional normalized descriptor action. This action is not an incremental modification of the preceding block. The action is denormalized to descriptor bounds, evaluated by the guarded surrogate, and converted to the next normalized target state and scalar reward. The 40-step horizon therefore functions as a fixed sequence of policy-training queries on a static surrogate response surface. Figure 2 summarizes this serialized query structure.

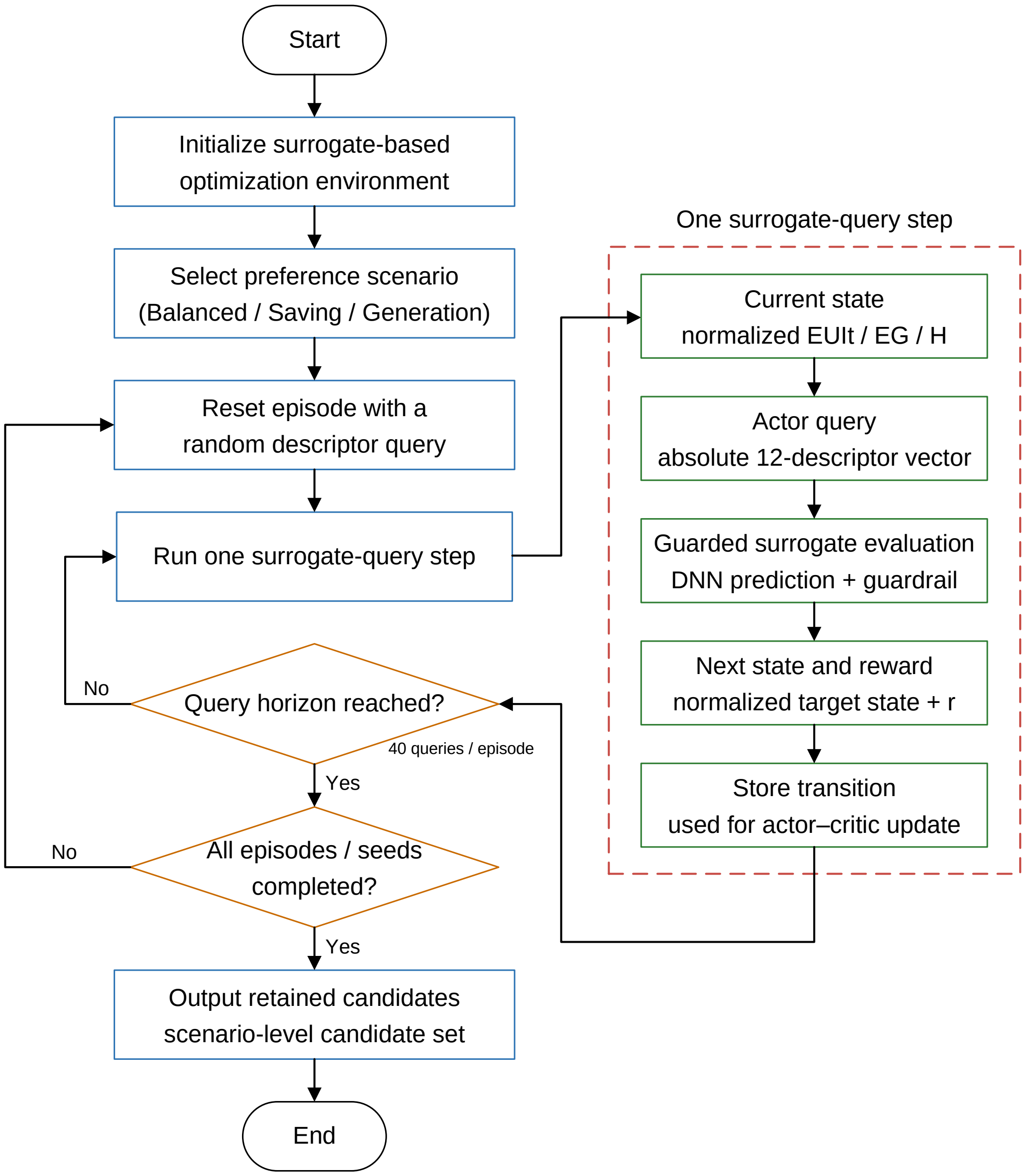

Figure 2: Serialized DDPG surrogate-query workflow. Each episode contains 40 absolute descriptor queries to the guarded surrogate environment.

The scalar reward is computed from normalized target distance to the preference-specific ideal direction. EUIt is minimized, while EG and H are maximized. Let y_i be the guarded surrogate output for target i, with $y_i^{\min}$ and $y_i^{\max}$ taken from the target bounds. The normalized target coordinate is

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}. \tag{1}$$

The ideal target vector is

$$\boldsymbol{u} = [0, 1, 1]^{\mathsf{T}}. \tag{2}$$

For preference weights $\boldsymbol{w}$, the weighted normalized distance is

$$d_w = \frac{\|\boldsymbol{w} \odot (\boldsymbol{z} - \boldsymbol{u})\|_2}{\|\boldsymbol{w}\|_2}. \tag{3}$$

The training reward is therefore

$$R = 1 - d_w. \tag{4}$$

The preference weights follow the target order [EUIt, EG, H]: Balanced uses $\boldsymbol{w} = [1/3, 1/3, 1/3]^\mathsf{T}$, Energy Saving uses $\boldsymbol{w} = [0.6, 0.2, 0.2]^\mathsf{T}$, and Energy Generation uses $\boldsymbol{w} = [0.2, 0.6, 0.2]^\mathsf{T}$.

This reward is separate from the fixed-domain post-hoc utility used later for benchmark comparison. The fixed-domain utility is used for post-hoc comparison, while R is the scalar signal used during serialized DDPG training. Figure 3 summarizes the actor-critic and target-network structure used for these surrogate-query updates.

Figure 3 summarizes the actor-critic update used in this surrogate-query environment. The state s is the normalized three-target vector, the action a is the normalized 12-descriptor query, r is the scalar reward, s_{next} is the next normalized target state returned by the guarded surrogate, and d is the episode-termination indicator. The online critic $Q(s, a)$ estimates the action value, while the target actor and target critic provide the delayed target used in the temporal-difference update. The critic minimizes the mean-squared error between $Q(s, a)$ and the TD target, while the actor is updated by maximizing the critic value of the actor-generated action.

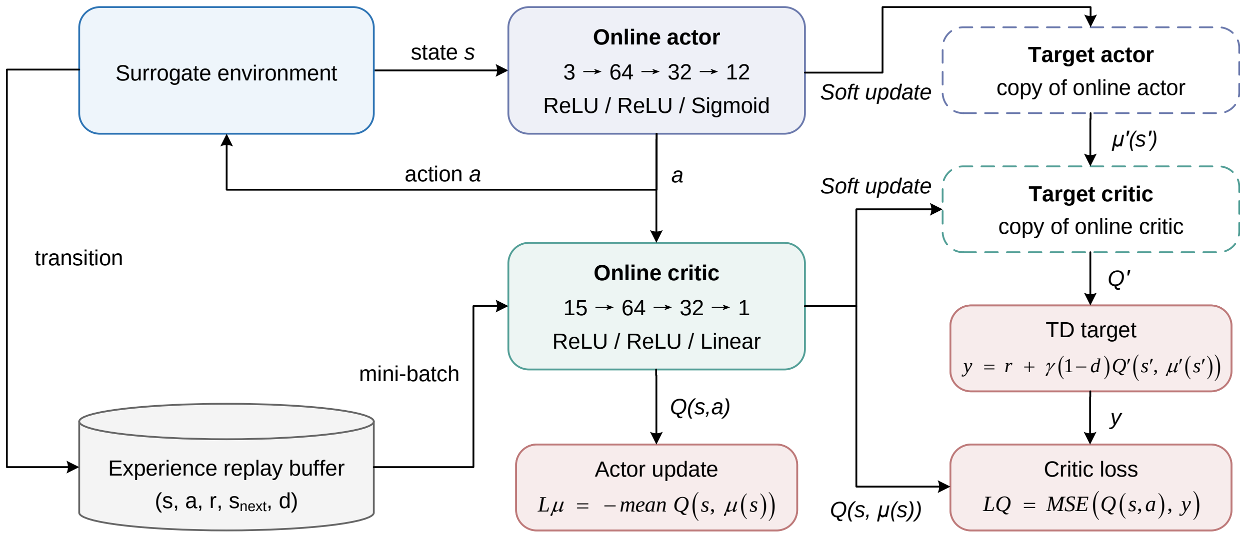

Figure 3: DDPG actor-critic architecture and target-network update structure used in the surrogate-query environment. The diagram identifies the normalized target state, 12-descriptor action, guarded surrogate transition, scalar reward, replay buffer, online actor and critic, target actor and critic, TD target, critic loss, actor update, and soft target-network update.

2.5. *Optimizer comparison protocol*

Table 4 summarizes the query budget and the retained candidates used for comparison. DDPG is evaluated through one best scalarized candidate from each seed, whereas NSGA-II retains the final population from each run. CMA-ES and RandomSearch are included as supplementary continuous-search references, with complete results provided in the Supplementary Information.

All optimizer families use 20 seeds or runs and 24,000 surrogate queries per seed or run. The main descriptor-space comparison between DDPG and NSGA-II is therefore matched at the query-budget level. Because the retained outputs differ in structure, archive metrics are interpreted together with equal-size summaries, feasible-block projection, and fixed-domain utility.

Table 4: Optimizer budgets and retained outputs.

Method	Runs	Queries/run	Retained rows	Retained output
DDPG	20 seeds	24,000	20	best scalarized point
NSGA-II	20 runs	24,000	2000	final populations
CMA-ES	20 runs	24,000	2000	supplementary archive
RandomSearch	20 runs	24,000	2000	supplementary archive

Complete CMA-ES and RandomSearch summaries are provided in Supplementary Section S3.

2.6. Feasible projection, physics-based cross-model stress test, and cross-climate sensitivity

Descriptor-space optimizer outputs are treated first as candidate descriptors. Each retained descriptor-space candidate is therefore mapped to the nearest generated feasible block in normalized descriptor space. This mapping quantifies projection distance, duplicate collapse, and changes in HV and IGD under the fixed normalized reference front. The projection step is a representation check that precedes morphology-level interpretation.

The limited physics-based cross-model stress test uses 24 locked cases. Eighteen direct feasible cases compare analytic response-generator values with physics-based outputs without optimizer projection, while six optimizer-linked cases decompose the path from descriptor-space candidate to nearest generated feasible block and then to physics-based result. The direct and optimizer-linked subsets answer different questions and are not merged into a single parity plot. The selected physics-based and climate-sensitive cases are implemented through a Grasshopper/Rhino-based workflow that links generated block geometry to Honeybee, EnergyPlus, and Radiance components. This workflow is used only for the locked case calculations in the physics-based cross-model stress test and cross-climate sensitivity analysis. The 2000-sample optimizer benchmark remains an analytic response-generation dataset, which provides the controlled response surface for comparing DDPG and NSGA-II under repeated surrogate queries. The baseline weather file used in the physical workflow serves as the reference condition for the cross-climate sensitivity analysis. The cross-climate sensitivity set then evaluates the same four block morphologies under three additional weather files representing Harbin, Beijing, and Guangzhou. The surrogate is not retrained, and the morphology set is not reselected after observing climate results.

3. Results

3.1. Descriptor coverage and surrogate assessment

The 2000 generated morphologies occupy a structured descriptor space rather than an unconstrained 12-dimensional hypercube. The PCA curve in Fig. 4 shows that six components are required to explain 95% of descriptor variance. This result indicates that the sample set preserves multi-dimensional variation while retaining geometric dependencies among descriptors. The dependency checks further show that FAR, BD, AF, and OSR are linked through the same footprint and gross-floor-area quantities. Consequently, descriptor-space search must be interpreted together with the feasible-block projection step.

The parity panels in Fig. 4 show that the selected DNN closely reproduces the analytic target surface within the sampled regime. Table 3 and Fig. 5 then extend this assessment beyond random folds by adding leave-one-OSLI-out, outer-shell, and feature-tail protocols. The grouped and boundary-oriented checks show modest error increases near the edges of the descriptor domain. This pattern is important for optimization because high-performing candidates tend to concentrate near target and descriptor limits, where average cross-validation accuracy alone is not sufficient to characterize surrogate behaviour.

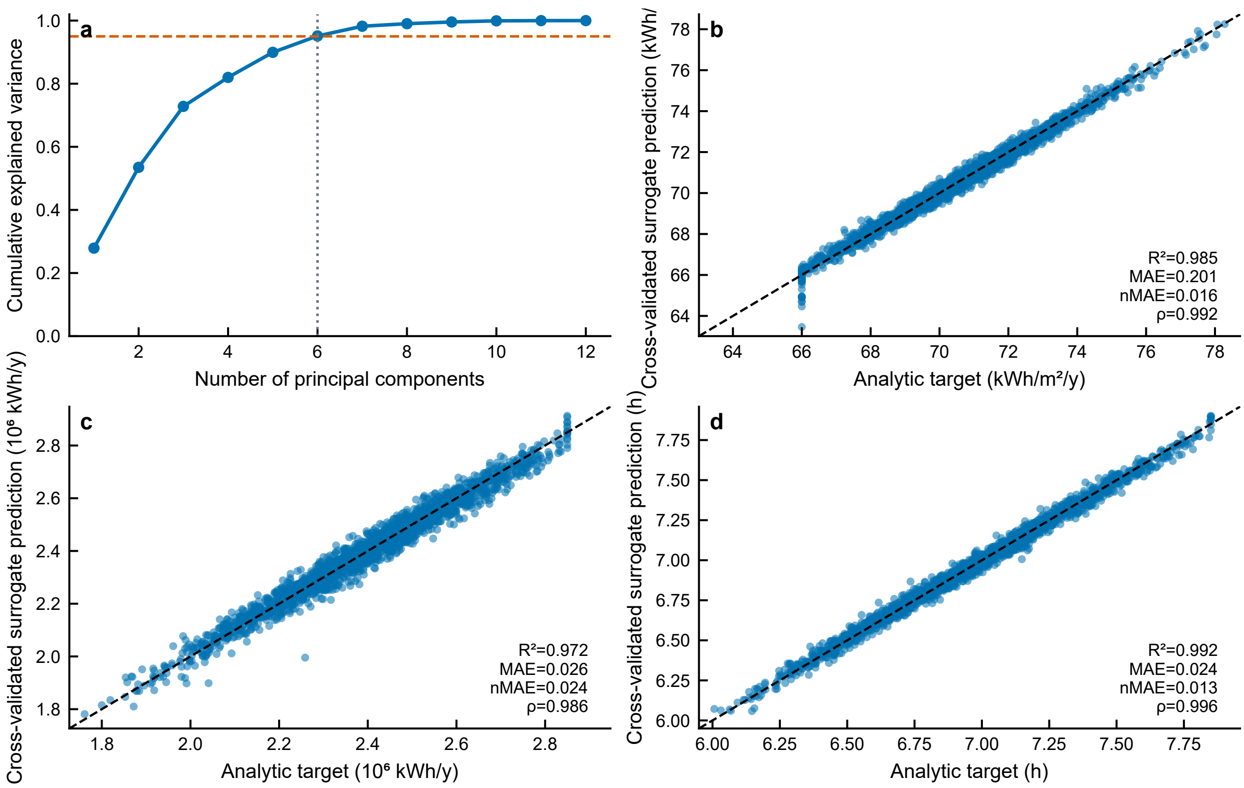

Figure 4: Descriptor coverage and cross-validated surrogate predictions.

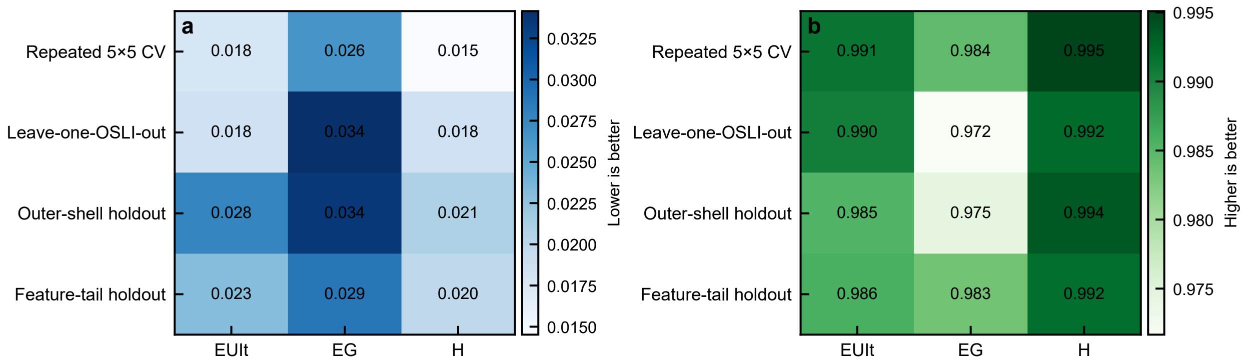

Figure 5: Surrogate assessment across random, grouped, and boundary-oriented validation regimes.

3.2. DDPG training behaviour

Figure 6 describes how the DDPG agents interact with the guarded surrogate during training. Across the three preference scenarios, rewards increase rapidly during early training and then enter a high-reward region with visible seed-to-seed variability. Because each episode contains a fixed sequence of 40 surrogate queries, the trajectories represent policy-training behaviour on a static response surface. They should therefore be read as search trajectories over descriptor queries, not as the physical evolution of one block design.

The later portions of the curves remain informative. The agents can repeatedly reach favourable surrogate regions, yet the best-to-final gap indicates that late-stage regression remains present. This behaviour is consistent with the retained-output strategy used in Table 4, where one best scalarized candidate is retained from each seed. The training figure therefore supports the use of DDPG as a preference-conditioned search comparator, while the final method comparison must still be evaluated through retained outputs and feasible-block projection.

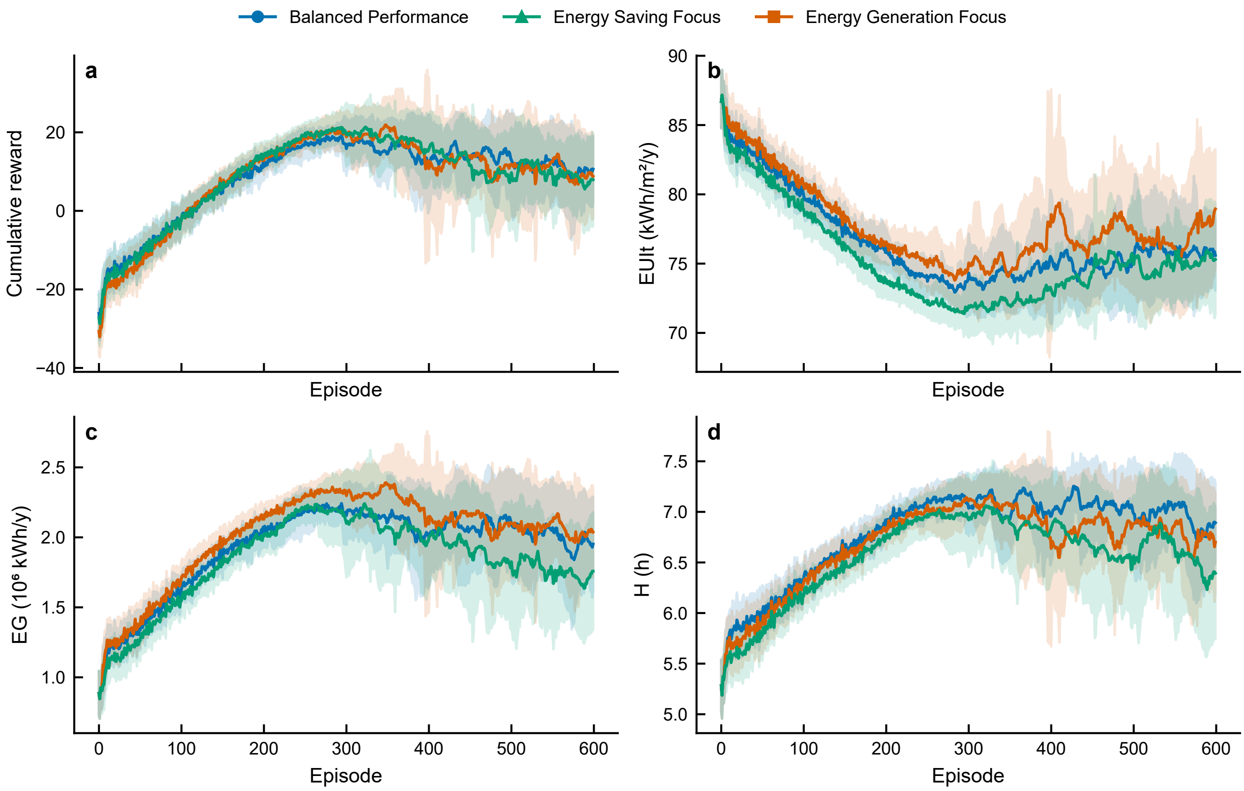

Figure 6: DDPG training trajectories across the three preference scenarios.

3.3. Descriptor-space optimizer comparison

Table 5 summarizes descriptor-space performance under the fixed normalized reference front, while Fig. 7 separates fixed-domain utility from equal-size archive metrics. NSGA-II is close to the HV ceiling under the fixed reference and remains the strongest method in the equal-size HV and IGD comparison. Among the three DDPG settings, the Generation-focused scenario gives the strongest descriptor-space archive metrics, followed by the Energy Saving and Balanced settings.

The comparison should be interpreted through the retained-output structure. DDPG contributes one best scalarized candidate from each seed, whereas NSGA-II contributes final populations from repeated runs. The equal-size summary reduces, but does not eliminate, this structural difference. Therefore, the descriptor-space result supports a method-comparison statement under the selected surrogate and fixed reference; it does not by itself describe feasible-block diversity. That question is addressed by the projection analysis in Fig. 8.

Table 5: Descriptor-space optimizer comparison using a fixed normalized reference.

Method	Scenario	Retained set	HV	IGD	Note
DDPG	Balanced	20 rows	0.661912	0.465522	scalarized output
DDPG	Saving	20 rows	1.002757	0.276529	scalarized output
DDPG	Generation	20 rows	1.170428	0.097551	scalarized output
NSGA-II	all	20 rows	1.330995	0.004710	equal-size subset
NSGA-II	all	2000 rows	1.330999	0.004947	full retained archive

All HV and IGD values use the same fixed normalized reference front. Complete CMA-ES, RandomSearch, and feasible-pool references are provided in Supplementary Section S3.

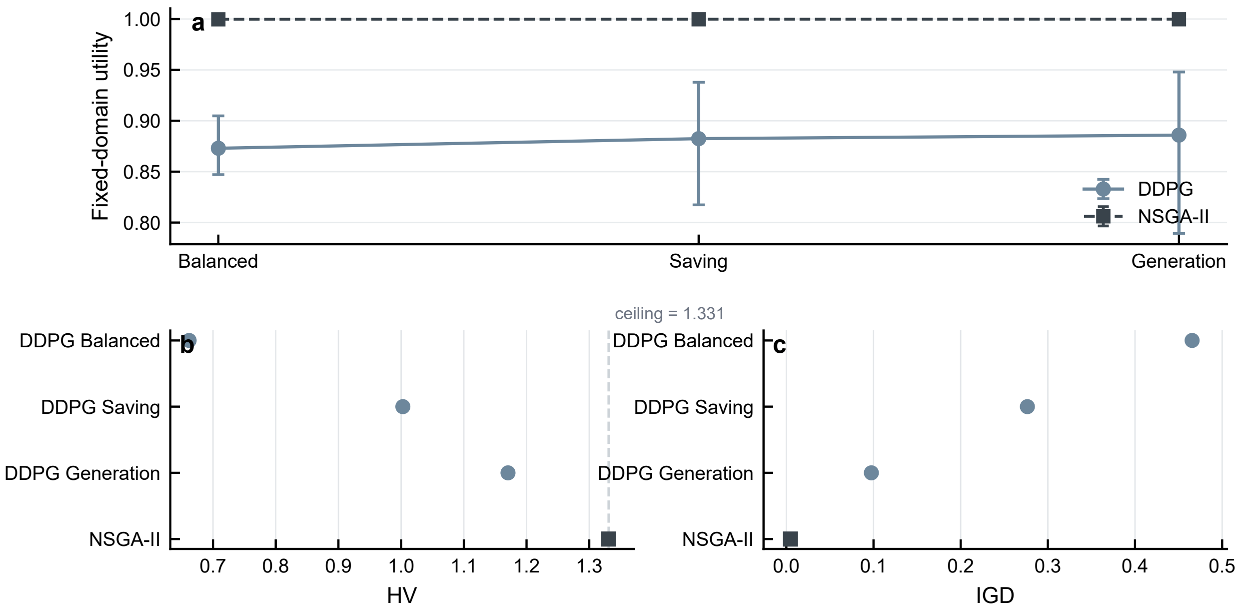

Figure 7: Descriptor-space DDPG and NSGA-II comparison under fixed-domain utility and equal-size retained outputs.

3.4. Projection to generated feasible morphologies

The projection analysis shows that descriptor-space coverage and feasible-morphology diversity are distinct. NSGA-II retains 2000 descriptor candidates, yet nearest-neighbour projection maps them to 51 unique generated feasible blocks. This many-to-one mapping is the central result of Fig. 8: a large descriptor archive can contract sharply when interpreted through the generated feasible morphology pool.

Projection distance adds a second layer of interpretation. The DDPG scenarios retain fewer descriptor candidates than NSGA-II, but their projected candidates are generally farther from the generated feasible blocks. In contrast, NSGA-II shows stronger descriptor-space coverage and shorter projection distances, yet its candidates still collapse to a much smaller feasible-block set. These results indicate that optimizer comparison should distinguish descriptor-space outputs, projected feasible morphologies, and the smaller subset that is later examined through physics-based calculations.

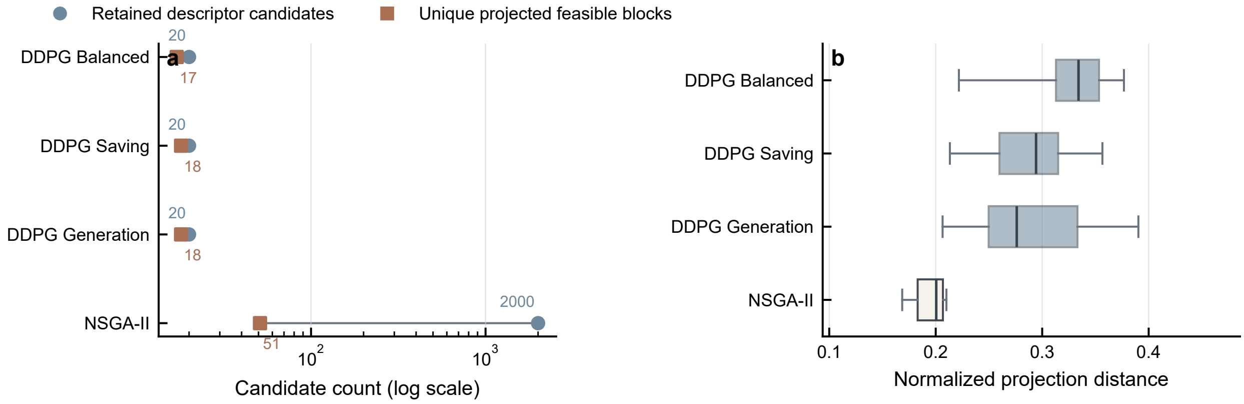

Figure 8: Descriptor-space candidate compression and projection distance after nearest-feasible-block mapping.

3.5. *Cross-model and cross-climate sensitivity*

The physics-based cross-model stress test completed all locked cases and provides a case-level comparison between analytic target values and physics-based outputs. Figure 9 uses only the 18 direct feasible blocks, so the parity plots are not affected by optimizer projection. The three targets show limited metric and rank consistency in this case set, with the weakest ordering stability appearing in the sunlight-related and demand-related comparisons. This result does not diminish the value of the stress test; instead, it clarifies the degree to which the analytic response surface and the physics-based calculations differ for selected generated blocks.

The climate analysis in Fig. 10 evaluates four selected blocks under three additional weather files relative to the baseline weather. The demand response is strongly climate-dependent, especially under the severe-cold Harbin weather. EG ordering remains stable across the additional climates, suggesting that the rooftop-generation proxy is less sensitive to the weather substitution for this four-block set. H ordering is less stable, which is consistent with its dependence on solar geometry, obstruction, and weather-specific direct-sun conditions. Table 6 summarizes these result boundaries and supports using the climate analysis as a limited sensitivity test rather than as a general cross-climate transfer claim.

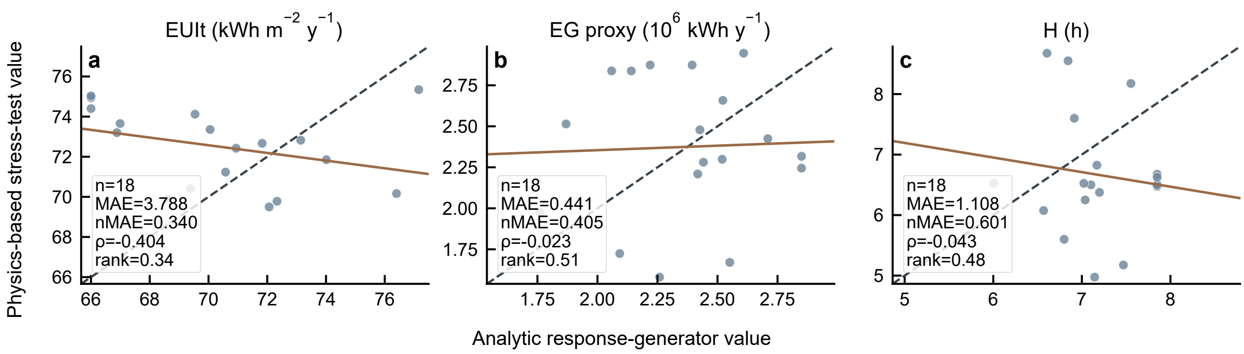

Figure 9: Cross-model stress test for 18 direct generated feasible blocks.

Table 6: Physical and climate sensitivity summary.

Assessment	Target or output	Scope	Interpretation
Cross-model stress test	EUIt	18 direct cases	limited consistency
Cross-model stress test	EG proxy	18 direct cases	proxy-based output
Cross-model stress test	H	18 direct cases	target-dependent response
Cross-climate sensitivity	EG ordering	four blocks, three climates	stable ordering
Cross-climate sensitivity	H ordering	four blocks, three climates	climate-dependent ordering
Cross-climate sensitivity	EUIt ordering	severe-cold Harbin case	climate-dependent ordering

The EG stress-test output is the GHI-based rooftop-PV proxy used in the locked physical case set.

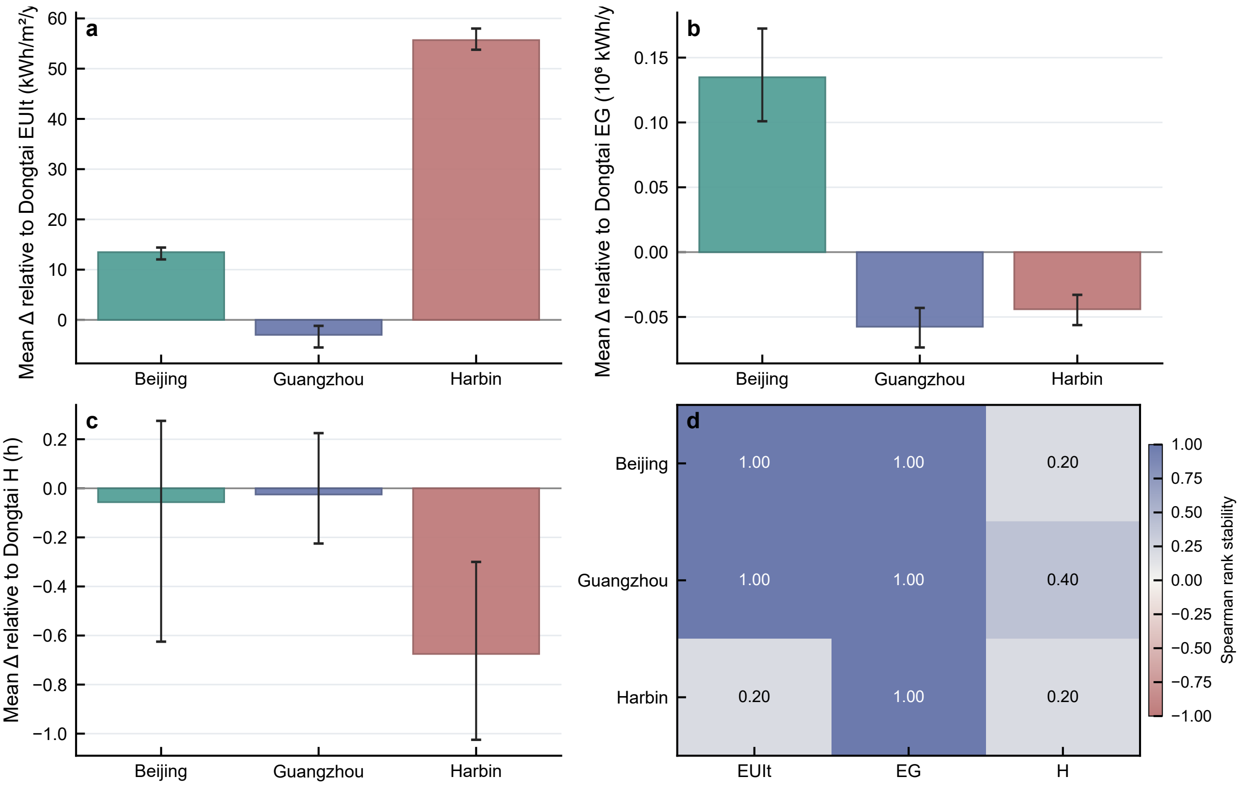

Figure 10: Cross-climate sensitivity of four selected blocks relative to baseline weather.

4. Discussion

4.1. Descriptor-space ranking and feasible morphology interpretation

The descriptor-space comparison indicates that NSGA-II provides stronger equal-size HV and IGD values than the three DDPG scenarios under the fixed normalized reference. This result is most informative as a comparison of search behaviour in the selected surrogate environment. Feasible-block interpretation requires the subsequent projection step because the optimizer outputs first exist as continuous descriptor-space candidates.

The projection analysis changes that interpretation. NSGA-II retains 2000 descriptor candidates, but these candidates map to 51 unique generated feasible blocks after nearest-neighbour projection. This compression shows that descriptor archive size and feasible-morphology diversity are different quantities. Projection distance and duplicate collapse are therefore necessary for interpreting continuous optimizer outputs in a morphology-generation workflow.

4.2. Physical and climate sensitivity boundaries

The physics-based cross-model stress test completed all 24 locked cases, but the agreement between analytic target values and physics-based outputs is limited. For the 18 direct feasible cases, Spearman correlation is negative or near zero for EUIt, the EG proxy, and H. These results indicate that descriptor-space ordering and physics-based ordering can diverge in the tested case set. The climate analysis adds a second boundary: EG ordering is stable in the four-block set, whereas H and severe-cold EUIt responses remain climate-dependent.

These two sensitivity analyses do not convert the descriptor-space comparison into a physical ranking result. They instead define where the analytic response surface, generated feasible blocks, and selected physical workflows agree only partially, which is the boundary needed for early-stage design-support interpretation.

4.3. Implications, limitations, and future work

For early-stage urban energy design, the workflow is useful because it separates fast descriptor-space screening from feasible-block projection and sensitivity analysis. A surrogate optimizer can identify promising descriptor regions, while projection and selected physics-based calculations help determine how those regions relate to generated block geometries. This separation makes the workflow more transparent for design-support use.

Several limitations remain. The optimizer comparison is conducted on an analytic response-generation dataset, while direct physical calculations are available for the locked stress-test cases. The 12 surrogate inputs are morphology-derived descriptors rather than independent unconstrained design variables. The DDPG horizon is best understood as a policy-training query sequence on a static surrogate. The cross-model stress test indicates limited rank consistency in the tested case set, and the climate analysis covers four blocks and three additional climates. Future work should connect optimization to a feasible morphology decoder, expand the physical simulation set, incorporate storage, EV charging, and distributed-energy interactions, and train climate-aware surrogates across multiple regions.

5. Conclusions

This study compared policy learning and Pareto search in a surrogate-assisted workflow for block-scale urban energy design. The workflow generates feasible residential block morphologies, converts them into 12 morphology descriptors, assigns EUIt, EG, and H through an explicit morphology-to-performance response surface, and trains a selected DNN surrogate from the 2000-sample regime of a nested scale study. Within this controlled descriptor-space setting, repeated validation shows low surrogate error, and NSGA-II reaches near-ceiling HV under the fixed normalized reference, while the three DDPG scenarios remain more dependent on preference setting and retained-output structure.

The main contribution is the representation-aware interpretation of the optimizer comparison. Projection compresses the 2000 NSGA-II descriptor candidates to 51 unique generated feasible blocks, showing that descriptor-space coverage and feasible-morphology diversity are distinct. The 24-case cross-model stress test completes all locked cases but indicates limited metric and rank consistency between analytic and physics-based outputs. The four-block cross-climate sensitivity analysis shows stable EG ordering but climate-dependent EUIt and H responses. Together, these results support an early-stage urban design workflow in which surrogate search, feasible-block projection, and sensitivity analysis are considered jointly before optimizer comparisons are used to guide block-scale urban energy decisions.

Data and code availability

The code, processed datasets, and result tables supporting the numerical analyses will be made available through a public branch of the project repository at https://github.com/haolinwang1313/DRL-Driven-Optimization. The public release will include the scripts required to regenerate the analytic response dataset, surrogate assessment summaries, optimizer comparison tables, feasible-projection results, physics-based stress-test summaries, and cross-climate sensitivity results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported in part by the Outstanding Youth Fund of Jiangsu Province under Grant No. BK20240076, and in part by the Excellent Young Scientists Fund, China, under Grant No. 52522704.

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